

| Dataset                          | $\#F_{total}$ | $\min( \text{Var}(F) )$ | $\max( \text{Var}(F) )$ | $\min( F )$ | $\max( F )$ |
|----------------------------------|---------------|-------------------------|-------------------------|-------------|-------------|
| Global                           | 4550          | 0                       | 8286433                 | 0           | 7689680     |
| $\hookrightarrow$ Lagniez [?]    | 1948          | 5                       | 229100                  | 10          | 399794      |
| $\hookrightarrow$ Soos [?]       | 1823          | 2                       | 8286433                 | 1           | 7689680     |
| $\hookrightarrow$ Plazar [?]     | 501           | 14                      | 486193                  | 31          | 2598178     |
| $\hookrightarrow$ Sundermann [?] | 278           | 0                       | 31012                   | 0           | 350120      |

Table 1: Dataset summary. The first column indicates the dataset, and the  $\#F_{total}$  column indicates how many satisfiable formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

| Dataset                          | $\#F_{total}$ | $\#D4$ | $\#\neg D4$ | $\#\text{CapKC} \wedge \neg D4$ | $\#\text{DivKC} \wedge \neg \text{CapKC} \wedge \neg D4$ |
|----------------------------------|---------------|--------|-------------|---------------------------------|----------------------------------------------------------|
| Global                           | 4550          | 185    | 670         | 80                              | 55                                                       |
| $\hookrightarrow$ Lagniez [?]    | 1948          | 53     | 213         | 40                              | 23                                                       |
| $\hookrightarrow$ Soos [?]       | 1823          | 100    | 393         | 40                              | 28                                                       |
| $\hookrightarrow$ Plazar [?]     | 501           | 31     | 61          | 0                               | 4                                                        |
| $\hookrightarrow$ Sundermann [?] | 278           | 1      | 3           | 0                               | 0                                                        |

Table 2: Experimental results regarding the scalability of DivKC and CapKC. Column  $\#F_{total}$  indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column  $\#\neg D4$  shows the number of formulae not compiled by D4. Column  $\#\text{CK} \wedge \neg D4$  shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC.

| Dataset                          | $\#F_{total}$ | $\#\neg D4$ | $\#\neg D4 \wedge \text{CapKC}$ | $\#\neg D4 \wedge \text{DivKC}$ | $\#\text{CapKC} \wedge \neg \text{DivKC} \wedge \neg D4$ |
|----------------------------------|---------------|-------------|---------------------------------|---------------------------------|----------------------------------------------------------|
| Global                           | 4550          | 670         | 80                              | 114                             | 21                                                       |
| $\hookrightarrow$ Lagniez [?]    | 1948          | 213         | 40                              | 52                              | 11                                                       |
| $\hookrightarrow$ Soos [?]       | 1823          | 393         | 40                              | 58                              | 10                                                       |
| $\hookrightarrow$ Plazar [?]     | 501           | 61          | 0                               | 4                               | 0                                                        |
| $\hookrightarrow$ Sundermann [?] | 278           | 3           | 0                               | 0                               | 0                                                        |

Table 3: Experimental results regarding the scalability of DivKC and CapKC.

| Dataset                          | $\#F$ | $Y_l \leq  R_F $ | $Y_h \geq  R_F $ | Coverage | $ R_{G_P}  \leq  R_F  \leq  R_{G_U} $ |
|----------------------------------|-------|------------------|------------------|----------|---------------------------------------|
| Global                           | 3670  | 0.969            | 0.857            | 0.826    | 1.000                                 |
| $\hookrightarrow$ Lagniez [?]    | 1689  | 0.969            | 0.895            | 0.864    | 1.000                                 |
| $\hookrightarrow$ Soos [?]       | 1307  | 0.969            | 0.906            | 0.875    | 1.000                                 |
| $\hookrightarrow$ Plazar [?]     | 403   | 0.968            | 0.789            | 0.757    | 1.000                                 |
| $\hookrightarrow$ Sundermann [?] | 271   | 0.967            | 0.487            | 0.454    | 1.000                                 |

Table 4: Experimental results for DivKC-AMC. Column  $\#F$  indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often  $|R_F|$  is within the confidence interval  $[Y_l; Y_h]$  and thus measures the accuracy of our method.

| Dataset                          | $\#F$ | $Y_l \leq  R_F $ | $Y_h \geq  R_F $ | Coverage | $ R_F  \leq  R_{G_U} $ |
|----------------------------------|-------|------------------|------------------|----------|------------------------|
| Global                           | 323   | 0.994            | 0.988            | 0.981    | 1.000                  |
| $\hookrightarrow$ Lagniez [?]    | 168   | 0.994            | 0.988            | 0.982    | 1.000                  |
| $\hookrightarrow$ Soos [?]       | 149   | 1.000            | 0.987            | 0.987    | 1.000                  |
| $\hookrightarrow$ Plazar [?]     | 4     | 1.000            | 1.000            | 1.000    | 1.000                  |
| $\hookrightarrow$ Sundermann [?] | 2     | 0.500            | 1.000            | 0.500    | 1.000                  |

Table 5: Experimental results for CapKC. Column  $\#F$  indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often  $|R_F|$  is within the confidence interval  $[Y_l; Y_h]$  and thus measures the accuracy of our method.

| Dataset                          | # F  | $Y_l \geq  R_{G_P} $ | $Y_h \leq  R_{G_U} $ | Both  | median( $r_c$ ) | max( $r_c$ ) |
|----------------------------------|------|----------------------|----------------------|-------|-----------------|--------------|
| Global                           | 3669 | 0.807                | 0.836                | 0.721 | 4.17e-9         | 0.54         |
| $\hookrightarrow$ Lagniez [?]    | 1689 | 0.844                | 0.869                | 0.760 | 1.8e-9          | 0.54         |
| $\hookrightarrow$ Soos [?]       | 1307 | 0.920                | 0.893                | 0.834 | 1.75e-7         | 0.0962       |
| $\hookrightarrow$ Plazar [?]     | 403  | 0.697                | 0.769                | 0.610 | 1.64e-13        | 0.0153       |
| $\hookrightarrow$ Sundermann [?] | 270  | 0.196                | 0.448                | 0.093 | 1.69e-13        | 0.281        |

Table 6: Experimental results comparing the bounds obtained with DivKC-AMC and with Lemma ?? . Column # F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have  $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$  and  $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$ . The last two columns represent the observed median and maximum values of the ratio  $r_c = \frac{\min(Y_h, |R_{G_U}|) - \max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$ , which was calculated exclusively if  $Y_l \leq |R_F| \leq Y_h$ . The number of formulae on which the last two columns are computed can easily be obtained by multiplying the # F column with the 'Coverage' column in Table 4.

| Dataset                          | # F | $Y_h \leq  R_{G_U} $ | median( $r_c$ ) | max( $r_c$ ) |
|----------------------------------|-----|----------------------|-----------------|--------------|
| Global                           | 334 | 0.994                | 0.00812         | 0.0257       |
| $\hookrightarrow$ Lagniez [?]    | 175 | 0.994                | 0.00753         | 0.0257       |
| $\hookrightarrow$ Soos [?]       | 153 | 0.993                | 0.00861         | 0.0257       |
| $\hookrightarrow$ Plazar [?]     | 4   | 1.000                | 0.00684         | 0.0257       |
| $\hookrightarrow$ Sundermann [?] | 2   | 1.000                | 0.00981         | 0.0155       |

Table 7: CapKC.

| Dataset                          | # F  | $l \leq Y_{ApproxMC} \leq h$ | $l \leq Y_{DivKC-AMC} \leq h$ | $l \leq Y_{CapKC-AMC} \leq h$ |
|----------------------------------|------|------------------------------|-------------------------------|-------------------------------|
| Global                           | 2713 | 0.973                        | 0.871                         | 0.981                         |
| $\hookrightarrow$ Lagniez [?]    | 1329 | 1.000                        | 0.889                         | 0.980                         |
| $\hookrightarrow$ Soos [?]       | 1177 | 0.964                        | 0.847                         | 0.979                         |
| $\hookrightarrow$ Plazar [?]     | 185  | 0.838                        | 0.892                         | 0.995                         |
| $\hookrightarrow$ Sundermann [?] | 22   | 1.000                        | 0.955                         | 1.000                         |

Table 8: CapKC.

| Dataset                          | # F  | # DivKC-AMC | $\log_{10}(\min)$ | mean   | median | $\log_{10}(\max)$ |
|----------------------------------|------|-------------|-------------------|--------|--------|-------------------|
| Global                           | 3188 | 2129        | -3.7              | 160.7  | 1.9    | 5.1               |
| $\hookrightarrow$ Lagniez [?]    | 1599 | 1160        | -3.0              | 239.1  | 2.1    | 4.9               |
| $\hookrightarrow$ Soos [?]       | 1349 | 855         | -3.6              | 6.2    | 1.8    | 3.6               |
| $\hookrightarrow$ Plazar [?]     | 218  | 97          | -3.7              | 1.3    | 0.9    | 1.2               |
| $\hookrightarrow$ Sundermann [?] | 22   | 17          | -1.0              | 5507.7 | 12.6   | 5.1               |

Table 9: DivKC vs Approxmc7.

| Dataset                          | # F  | # CapKC-AMC | $\log_{10}(\min)$ | mean   | median | $\log_{10}(\max)$ |
|----------------------------------|------|-------------|-------------------|--------|--------|-------------------|
| Global                           | 2514 | 1554        | -2.3              | 179.1  | 1.8    | 5.0               |
| $\hookrightarrow$ Lagniez [?]    | 1205 | 773         | -2.3              | 287.9  | 2.0    | 5.0               |
| $\hookrightarrow$ Soos [?]       | 1111 | 651         | -2.2              | 3.3    | 1.6    | 2.8               |
| $\hookrightarrow$ Plazar [?]     | 176  | 111         | -1.2              | 2.1    | 1.5    | 1.1               |
| $\hookrightarrow$ Sundermann [?] | 22   | 19          | -0.9              | 4517.2 | 11.1   | 5.0               |

Table 10: CapKC vs Approxmc7.

| Dataset                          | # F | # CapKC-AMC | $\log_{10}(\min)$ | mean  | median | $\log_{10}(\max)$ |
|----------------------------------|-----|-------------|-------------------|-------|--------|-------------------|
| Global                           | 421 | 182         | -2.2              | 250.3 | 0.6    | 5.0               |
| $\hookrightarrow$ Lagniez [?]    | 207 | 92          | -2.2              | 506.3 | 0.6    | 5.0               |
| $\hookrightarrow$ Soos [?]       | 194 | 74          | -2.2              | 2.7   | 0.5    | 2.2               |
| $\hookrightarrow$ Plazar [?]     | 16  | 12          | -0.9              | 2.8   | 2.1    | 0.7               |
| $\hookrightarrow$ Sundermann [?] | 4   | 4           | 0.0               | 5.6   | 6.6    | 0.9               |

Table 11: CapKC vs Approxmc7.